Failure of the discrete maximal-variation conjecture

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Task 1. Resolve the original conjecture. For a function on the integers, use the centered maximal function and adjacent-difference variation

$$Mf(n) = \sup_{r \in \mathbb{Z}_{\geq 0}} \frac{\sum_{k=-r}^{r} |f(n+k)|}{2r+1}, \qquad \mathrm{Var}_p(f)^p = \sum_{n \in \mathbb{Z}} |f(n+1) - f(n)|^p.$$

Conjecture 1, equation (5.4), of González-Riquelme and Madrid [1] proposes, for $1/2 < p \leq 1$ and $f \in \ell^p(\mathbb{Z})$,

$$\mathrm{Var}_p(Mf) \leq C_p \, \mathrm{Var}_p(f), \qquad C_p^p = \sum_{j=0}^{\infty} \left(\frac{2}{(2j+1)(2j+3)} \right)^p. \tag{1}$$

This is false for every $1/2 < p < 1$. We first show that no finite replacement constant works in that range, then give one fixed three-point example refuting the specific proposed constant for all those exponents.

Theorem 1. *For each $1/2 < p < 1$,*

$$\sup_{\substack{f \neq 0 \\ f \text{ finitely supported}}} \frac{\mathrm{Var}_p(Mf)}{\mathrm{Var}_p(f)} = \infty.$$

Proof. Let $N \geq 1$ be an integer and put $f_N = \mathbf{1}_{[-N,N] \cap \mathbb{Z}}$. There are exactly two unit jumps, so $\mathrm{Var}_p(f_N)^p = 2$. We claim that

$$Mf_N(q) = \begin{cases} 1, & |q| \leq N, \\ (2N+1)/(2|q| + 2N + 1), & |q| > N. \end{cases} \tag{2}$$

For $q > N$, the first radius capturing j support points is $q - N + j - 1$, with average

$$\frac{j}{2q - 2N + 2j - 1}, \qquad 1 \leq j \leq 2N+1.$$

These fractions strictly increase with j, because $2q - 2N - 1 > 0$. Once the whole support is captured, every larger radius decreases the average. This proves the claim; symmetry and radius zero handle the other centers.

For $N + 1 \leq q \leq 2N$, formula (2) gives

$$Mf_N(q) - Mf_N(q+1) = \frac{2(2N+1)}{(2q + 2N + 1)(2q + 2N + 3)} \geq \frac{4}{63N}.$$

Indeed, the numerator is at least $4N$, and the denominator is at most $(6N+1)(6N+3) \le 63N^2$. There are N such jumps on each side. Consequently

$$\left(\frac{\mathrm{Var}_p(Mf_N)}{\mathrm{Var}_p(f_N)}\right)^p \ge N\left(\frac{4}{63N}\right)^p = \left(\frac{4}{63}\right)^p N^{1-p} \longrightarrow \infty.$$

Each f_N belongs to ℓ^p, and each output variation is finite: for fixed N, the tail jumps are $O(q^{-2})$ and $2p > 1$. □

A fixed counterexample to the proposed constant. Take $f = f_1 = \mathbf{1}_{\{-1,0,1\}}$. Its maximal function is

$$Mf(n) = 1 \quad (|n| \le 1), \qquad Mf(n) = \frac{3}{2|n|+3} \quad (|n| \ge 2).$$

The jump sizes on either side are

$$a_0 = \frac{4}{7}, \qquad a_j = \frac{6}{(2j+5)(2j+7)} \quad (j \ge 1).$$

Set also

$$b_j = \frac{2}{(2j+1)(2j+3)} \quad (j \ge 0).$$

Both a and b are decreasing positive sequences of total mass one. For $m \ge 1$, their partial sums are

$$A_m = \sum_{j<m} a_j = 1 - \frac{3}{2m+5}, \qquad B_m = \sum_{j<m} b_j = 1 - \frac{1}{2m+1},$$

so

$$B_m - A_m = \frac{2(2m-1)}{(2m+1)(2m+5)} > 0. \tag{3}$$

We make the infinite-sequence concavity step explicit. For any decreasing probability sequence $z = (z_j)$, put

$$K_z(t) = \sum_j \min(z_j, t) \qquad (t > 0).$$

If $Z_m = \sum_{j<m} z_j$ and $Z_0 = 0$, then

$$\sum_j (z_j - t)_+ = \max_{m \ge 0}(Z_m - mt), \qquad K_z(t) = 1 - \sum_j (z_j - t)_+.$$

The maximum is attained at the finite number of entries exceeding t. Equation (3) therefore implies $K_a(t) \ge K_b(t)$. This inequality is strict for $4/7 < t < 2/3$: every $a_j < t$, so $K_a(t) = 1$, while $b_0 = 2/3 > t$ gives $K_b(t) < 1$.

For $0 < p < 1$, direct integration gives

$$z^p = p(1-p)\int_0^\infty \min(z,t)t^{p-2}\,dt.$$

Tonelli's theorem and the preceding strict interval now show

$$\sum_j a_j^p > \sum_j b_j^p \qquad (1/2 < p < 1).$$

Both sums are finite because their terms are $O(j^{-2p})$. Finally,

$$\left(\frac{\mathrm{Var}_p(Mf)}{\mathrm{Var}_p(f)}\right)^p = \frac{2\sum_j a_j^p}{2} > \sum_j b_j^p = C_p^p.$$

Thus the same three-point function refutes (1) for every exponent in the open interval.

Scope. At $p = 1$ both jump sums equal one, so these examples give equality. The proof refutes the universally quantified conjecture and all finite variation-to-variation constants for $1/2 < p < 1$; it does not resolve the general endpoint $p = 1$. The MathDB strings $p \neq (1/2, 1]$ and the malformed ℓ^p are transcription defects: the source and the record's explanatory sentence fix the intended range and space. The source's proved ℓ^p-to-variation bound is a different estimate and is not contradicted.

Task 2. Quantify the dependence on support size. Determine the sharp growth of the best variation-to-variation constant when the input is supported on an interval of at most L lattice points, for fixed $1/2 < p < 1$.

Task 3. Resolve the endpoint variation bound. Prove or refute $\mathrm{Var}_1(Mf) \leq \mathrm{Var}_1(f)$ for general $f \in \ell^1(\mathbb{Z})$, extending beyond the already known indicator-function case.

Task 4. Study radius truncation. For the maximal operator restricted to radii $r \leq R$, determine the optimal variation constant and its growth as $R \to \infty$ when $1/2 < p < 1$.

Task 5. Find a replacement estimate. Determine sharp bounds combining input variation with a measure of support size, mass or decay that controls the long-block obstruction while retaining the established ℓ^p-to-variation inequality.

References

[1] C. González-Riquelme and J. Madrid, *Sharp inequalities for maximal operators on finite graphs, II*, J. Math. Anal. Appl. **506** (2022), 125647, doi:10.1016/j.jmaa.2021.125647; arXiv:2011.02630, Conjecture 1.